

AllerGenie



AllerGenie Team Introductions



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Problem Space



Daily Challenges



Limited Restaurant Options

Fear of cross-contaminations



Endless Label Reading

Time-consuming ingredient checks



Grocery Shopping Anxiety

Complex ingredients verification

Impact on Daily Life

These challenges do not just restrict diet – they restrict life, travel, and daily activities for **33 million Americans**.



Eight Most Common Allergies

According to the Food and Drug Administration



EGGS



SOY



MILK



WHEAT



PEANUTS



TREE NUTS



SHELLFISH



FISH

These allergies are prevalent across many foods and cuisines



SARAH

Customer journey map

A 34-year-old working mother, juggles her career with raising a family. She is highly organized but often feels time-constrained. Her son's severe peanut allergy influences many of her choices, especially related to meal planning and grocery shopping.



Key Activities



Phase 1
Easy Web
Access

Seamless Onboarding

Easy access, fast loading, intuitive design—no access barriers.



Phase 2
Select
Allergens

Allergens & Dietary Configuration

Easy selection and management of allergens and preferences without account creation.



Phase 3
Barcode
Scanning

Effortless Scanning

Accurate scanning, fast response, clear issue handling, seamless product ID.



Phase 4
Allergen
Alerts

Clear and Accurate Alerts

Accurate detection, rapid response—users feel reassured.



Phase 5
Recipe
Generation

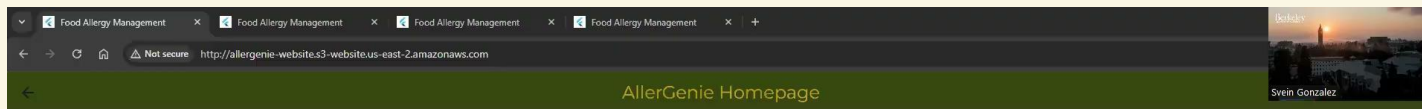
Tailored Meal Suggestions

Personalized, accurate recipe suggestions—easy to follow with allergen-safe alternatives.



Demo

Live Demo/Recorded Demo



Welcome to AllerGenie!



Check for Allergens in Your Favorite Foods!



Hungry for an Allergy Friendly Meal?





Data & Pipeline

Datasets

Allergy Detection



3.4 million products from OpenFoodFacts

A total of **656K products**, representing **19%** of the collection, are used in the allergy detection feature.



Limitations and Potential Challenges:

1. The inclusion of only U.S. based products
2. ~ 88% of U.S. products have missing values in the allergens column

Sample Data Record for **Nutella Mini Cups**

Code	0009800801107
URL	http://world-en.openfoodfacts.org/product/0009800801107/nutella-mini-cups
Product Name	Nutella Mini Cups
Brands	Nutella, Ferrero
Countries Tags	United States
Ingredients	Sugar, palm oil, hazelnuts, cocoa, skim milk, whey (milk), lecithin as emulsifier (soy), vanillin: an artificial flavor
Allergens	Milk, Nuts, Soybeans



Data Pipeline: Allergy Detection

Step 1: Barcode Scanning



User uploads an image of a product's barcode



Barcode image processed using OpenCV to convert to grayscale



Use pyzbar to decode the barcode into numeric values

Step 2: Database Lookup



The app checks the decoded barcode against OpenFoodFacts dataset (xlsx.)



A manually defined dictionary maps allergens to possible ingredient terms

Step 3: User Input



The app retrieves user input(s) for allergies they want to check

Step 4: Safety Evaluation



If a match is found, the app flags the product as **unsafe**.

Datasets

Recipe Generation



74,465 unique recipes from Hugging Face with no missing values

A total of **15K recipes**, representing **20%** of the collection, are stored in our RAG pipeline, forming a well-balanced recipe knowledge base. Recipes are stored in **AWS S3 buckets as csv files**, enabling optimized retrieval.



Wide-ranging diversity in culinary styles and allergy types

Cuisine Type	# of Recipes Note: A recipe may belong to multiple cuisine types
American	52K (69%)
Italian	15K (20%)
French	8.9K (12%)
European	5.2K (7%)
Other: Mexican, Mediterranean, Asian, Southern, Middle Eastern, British, Indian, Caribbean, Chinese, Spanish, Japanese...	

Allergy Type	# of Recipes Note: A recipe may contain multiple allergies.
Egg	20K (27%)
Milk	20K (27%)
Nut	17K (23%)
Dairy	17K (22%)
Wheat	13K (18%)
Other: Meat, Fish, Pepper, Spice, Cheese, Garlic, Grain, Cow's Milk, Onion, Peanuts	

Data Pipeline: Recipe Generation

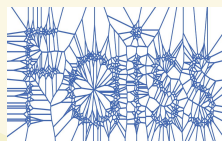
Step 1: Build Vector Store



Recipes Datasets
stored in S3 Buckets



Embedding Model bge-large-en-v1.5



FAISS Index

Step 2: Get Embeddings for User Query



Embedding Model bge-large-en-v1.5

Step 3: Retrieve Relevant Recipes



Step 4: Pass prompt with query and relevant documents to LLM

Query
+
Prompt
+
Retrieved Docs



LLM Model
Meta-Llama-3-8B-Instruct

Step 5: Extract titles from recipe outputs and generate images

`/ ( regex ) /`

Title Extraction

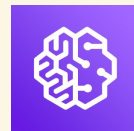


Image Generation Model
stable diffusion 2.1

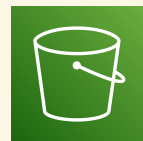


Image Storage in s3 Buckets



Evaluating: LLM Outputs

Five Evaluation Scenarios

We wanted to challenge the LLM with difficult allergy scenarios...



MARIA

Allergy



PEANUTS

NUTS

Wants to Cook

Thai Curry



JASON

Allergy



EGGS

MILK

Wants to Cook

Brownies



MICHAEL

Allergy



FISH

SHELLFISH

Wants to Cook

Japanese Dumplings



NORA

Allergy



WHEAT

Wants to Cook

Italian Chicken
Parmesan



FATIMA

Allergy

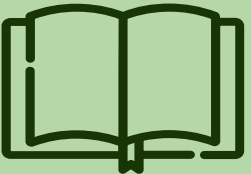


SOY

Wants to Cook

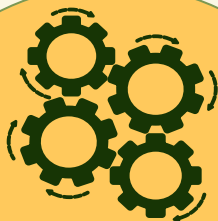
Vegan Tacos

Evaluation Metrics

	20%	20%	20%	40%
				
FOCUS	EFFECTIVENESS OF MODEL		READABILITY	ALLERGEN SCORE
METRIC USED	BLEU SCORE (Precision)	ROUGE SCORE (Recall)	FLESCH SCORE	1 / (ALLERGIES REFERENCED + 1)
COMMENT	LLM OUTPUT COMPARED TO "GOLD ANSWER"		QUANTIFIES READABILITY	ALLERGEN LISTS FROM JOHN HOPKINS

SCORES WERE WEIGHTED FOR FIVE DIFFERENT SCENARIOS

Parameters Fine-tuned



Model & Model Features

Evaluated Language Learning Models, embeddings, temperature settings, and top_p

High Impact



Prompt Engineering

Experimented with different queries and prompt formats

Very High Impact



Documents Used

Different document sizes, chunk and chunk overlaps

Low Impact

Model Evaluation Results

Optimized Model Output

Model	Llama3 8b	Mistral 7B	Mistral 7B-Instruct
BLEU Score (20%)	0.4016	0.3544	0.2144
Rouge Score (20%)	0.5497	0.4994	0.3414
Readability Score (20%)	0.5792	0.6242	0.6176
Allergen Score (40%)	1.0000	0.9000	1.0000
Final Score	0.7061	0.6556	0.6347



Highest Performing Combination

Scenario 3: Create a recipe for Japanese Dumplings without *FISH* or *SHELLFISH*

Generated Output

Pork Dumplings with Soy-Ginger Sauce

Ingredients:

- 1 pound pork butt, coarsely ground
- ...

Soy-Ginger Sauce (recipe follows)

- ...
- 1 tablespoon peeled, julienned gingerroot, soaked in ice water

Instructions:

1. In a large bowl, combine the pork, scallions, soy sauce, sesame oil, and gingerroot. Mix well so that the flavors will penetrate the meat.
2. Then add the cabbage and mix until all the ingredients are thoroughly combined.
3. Place 1 dumpling wrapper on a plate or a board. Place 1 scant tablespoon of pork mixture in the center.
4. Fold the wrapper over the filling to form a half-moon shape, and press the edges together to seal the dumpling.
5. Repeat with the remaining wrappers and filling.
6. Cook the dumplings in a large pot of boiling water for 5-7 minutes, or until they float to the surface.
7. Serve with Soy-Ginger Sauce.

Note: You can also pan-fry the dumplings with a small amount of oil until golden brown on the bottom **before boiling**.

Gold Answer

Pork Dumplings with Soy Ginger Sauce

Ingredients:

- 1 pound pork butt, coarsely ground
- ...
- 1 tablespoon peeled, julienned ginger root, soaked in ice water

Instructions:

1. In a large bowl, combine pork, scallions, soy sauce, sesame oil, and ginger root. Mix well.
2. Add cabbage and mix until all ingredients are thoroughly combined.
3. Place 1 dumpling wrapper on a plate or board. Place 1 scant tablespoon pork mixture in the center.
4. Moisten the edges of the wrapper with water and fold into a half moon shape.
5. Stand the dumplings upright and pleat the sides.
6. Repeat steps 3-5 with the remaining dumpling wrappers and pork filling.
7. To pan fry dumplings, heat corn oil in a large non-stick skillet. Cook dumplings over low heat until golden and crisp.
8. To boil dumplings, fill a large stockpot with water and bring to a boil. Add dumplings and cook until they float. Taste to check doneness.
9. Combine all ingredients in a small bowl and mix thoroughly. Serve.
10. To make sauce, combine all ingredients in a small bowl and mix thoroughly. Keep in the refrigerator up to 1 week without ginger root. Add ginger root when ready to serve.

BLEU Score	Rouge Score	Readability	Allergen Check	Total Score
0.5906	0.7411	0.5695	1.0000	0.7800

Examples of Poor Performing Outputs

Extra Sections

Lower scoring recipes included extra sections:

- Title
- **Allergen Information**
- **Cuisine Authenticity**
- **Recipe Differentiation**
- Ingredients
- **Allergen-Safe Substitutions**
- **Cross-Contamination Prevention**
- Instructions
- **Servings & Storage**
- **Final Safety Check**

Includes the Allergen

In one nut-free recipe, the following items were included:

- 1/2 cup macadamia nuts (omitted)
- 1 teaspoon shrimp paste (optional, omit for nut-free)
- Peanut oil

Peanut oil should have been removed to be extra safe, and the nuts should have been automatically removed from the recipe.

The flag for nuts on shrimp paste was not required.



Evaluating: A.I. Images

Evaluating Generative A.I. Images

Easy Pad Thai Recipe Without Peanuts



Real Image



AI Image

How well does the Artificial Intelligence (AI) image reflect the real image and label?

Very Bad Representation

Very Good Representation

0	1	2	3	4	5	6	7	8	9	10
☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

Experiment Overview

- An A/B Experiment was conducted with 120 participants.
- Respondents were asked to rate five A.I images on a scale of 0 to 10 (Bad → Good).
- Two variables were adjusted in the survey:
 - Model (1.5 vs. 2.1)
 - Prompt (Title vs. Recipe)
- Respondents were randomly shown an image for each of the five scenarios.
- Highest scoring combination was used in our model.

Highest performing image model...

★ Stable Diffusion 2.1 Model with a Short Prompt ★

	Dependent variable:			
	Respondent Score (0 to 10)			
	Model Only (1)	Prompt Only (2)	Model and Prompt (3)	Interaction Added (4)
Model 2.1	0.272 (0.222)	Positive & NOT Statistically Significant	0.263 (0.222)	0.339 (0.308)
Short Prompt		0.468** (0.234)	0.463** (0.222)	0.538* (0.313)
Model / Prompt Interaction				Positive & Statistically Significant -0.156 (0.443)
Constant	5.460*** (0.157)	5.362*** (0.202)	5.238*** (0.189)	5.202*** (0.217)
Observations	600	600	600	600
R ²	0.002	0.007	0.010	0.010
Adjusted R ²	0.001	0.006	0.006	0.005
Note:				*p<0.1; **p<0.05; ***p<0.01

Our number of observations (5 x 120) exceeds the power requirements to calculate significance

Project Summary



1

Combined image generation model with RAG system for the recipe generation

2

Platform combined two features to cover broader spectrum of food preparation

3

Leveraged AWS SageMaker to facilitate collaboration



User Feedback on Website

23 Respondents



100%

Generated recipes
excluded allergens

100%

Relevant recipes

73%

Users had trouble
scanning barcode *

* Prior to coding update

Aspirational Roadmap



Add Dietary Preferences, Expand Products Database

Based on user feedback



Link Two Features

Ability to generate recipe with the scanned product



Improve Image Generation

Explore additional image generation models

Project Mission



“Turning **food fear** into **food freedom**,
one meal at a time”

Q&A

